Ramil Khafizov

PhD Researcher (2024–2027) | Generative Models | Visual Foundation Models | 3D Vision

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Selected Publications

NAMVIS: Next-Scale Autoregressive Multi-View Image Synthesis
Ramil Khafizov, Ilya Statsenko, Ruslan Rakhimov, Artem Komarichev, Peter Wonka, Evgeny Burnaev.
NeurIPS 2026 [Paper] [Project] [Code]

One Algorithm to Align Them All
Boyi Pang, Savva Ignatyev, Vladimir Ippolitov, **Ramil Khafizov**, et al.
CVPR 2026 [Paper] [Project]

Preprints & Submissions

G-CUT3R: Guided 3D Reconstruction with Camera and Depth Prior Integration
Ramil Khafizov, Artem Komarichev, Ruslan Rakhimov, Peter Wonka, Evgeny Burnaev.
arXiv preprint, 2025 [Paper]

ARGeoSplat: Autoregressive Geometry Generation and Refinement for Sparse-View 3D Reconstruction
Ramil Khafizov, Artem Komarichev, Ilya Statsenko, Ruslan Rakhimov, Peter Wonka, Evgeny Burnaev.
Submitted to ICLR 2027

Research Experience

Applied AI Institute
Deep Learning Researcher *Sep 2024 – Present*

- **Generative model research (NAMVIS):** Led end-to-end development of a next-scale autoregressive model that generates geometrically consistent novel views in parallel; introduced multi-scale camera-geometry conditioning (ProPE) in self- and cross-attention. Achieved 21.77 dB mean PSNR versus 18.57 dB for the strongest evaluated diffusion baseline, with over 3× faster inference.
- **Geometric foundation model adaptation (G-CUT3R):** Developed flexible depth and camera-prior conditioning for feed-forward 3D reconstruction using modality-specific encoders and zero-initialized fusion. Reduced reconstruction accuracy error by up to 62% against baselines.
- **Joint appearance–geometry generation (ARGeoSplat):** Extended next-scale autoregression to generate RGB views and quantized 3D pointmaps jointly, with decoded-geometry supervision and cost-volume depth refinement for sparse-view 3D Gaussian reconstruction.
- **Training-free aligned generation (ATATA):** Contributed to the design and evaluation of a training-free rectified-flow method for aligned image, video, and 3D generation; fine-tuned diffusion LoRAs for related experiments.
- **Large-scale training:** Designed and ran training experiments for generative and reconstruction models up to 2B parameters; used multi-node setups of up to 32 H100 GPUs.
- **End-to-end research systems:** Independently built rendering, camera processing, dataset filtering, and WebDataset streaming pipelines for roughly 220K NAMVIS objects and up to 400K ARGeoSplat scenes; resolved training-memory, throughput, and stability bottlenecks.
- **Open-source flow matching model reproduction:** Independently implemented, trained, and released a PyTorch reproduction of Shortcut Models, demonstrating improved one-/few-step generation over standard flow matching. [Code]

Artec 3D artec3d.com

Deep Learning Researcher *Sep 2021 – Sep 2024*

- **Generative 3D editing:** Developed NeRF and 3D Gaussian Splatting editing methods using frozen 2D diffusion models, iterative image–3D optimization, and segmentation-guided inpainting.
- **Spatial vision-language modeling:** Adapted VLMs with LoRA to predict structured scene graphs, object relationships, and viewpoint-dependent spatial relations.
- **Promptable 3D segmentation:** Distilled CLIP image features into 3D Gaussian Splatting representations, with contrastive regularization for semantic separation.
- **Research-to-product transfer:** Developed 3D and computer vision algorithms incorporated into commercial products and reusable internal libraries.

Technical Skills

Generative modeling: Autoregressive models, diffusion, flow matching, rectified flow, geometry-aware attention.

3D computer vision: 3D reconstruction, novel-view synthesis, neural rendering, NeRF, 3D Gaussian Splatting.

Frameworks & training: PyTorch, PyTorch Lightning, DDP/FSDP, multi-node training, WebDataset, LoRA.

Programming & systems: Python, C/C++, CUDA, Linux, Bash, Docker.

Education

Applied AI Institute Sep 2024 – Sep 2027

PhD in Computer Science (in progress)

Innopolis University Aug 2019 – Jul 2021

MSc in Computer Science (Robotics and Computer Vision), GPA 4.82/5.00, with honors

Kazan Federal University Sep 2015 – Jun 2019

BSc in Engineering, GPA 4.50/5.00

Selected Achievement

Hilti SLAM Challenge 2023: 3rd place (Team Strelka). [Leaderboard]